

Problem Sheet 9

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1. Consider the following set of sentences $\Gamma = \{F_1, F_2, F_3, F_4\}$ such that

$$F_1 = \forall x (\forall y (C(x, y) \rightarrow R(y)) \rightarrow H(x))$$

$$F_2 = \forall x (G(x) \rightarrow R(x))$$

$$F_3 = \forall x (\exists y (C(y, x) \wedge G(y)) \rightarrow G(x))$$

$$F_4 = \neg \forall x (G(x) \rightarrow H(x)).$$

- (a) What is the signature τ of Γ ?
 - (b) Skolemize $F_1, \dots, F_4$ and obtain $G_1, \dots, G_4$. What is the signature of $G_1, \dots, G_4$?
 - (c) Show that propositional resolution gives $\emptyset$ by resolution applied on ground instances of $G_1, \dots, G_4$.
2. Give an example of a finite set of clauses F in first-order logic such that $Res^*(F)$ is infinite.
3. Give an example of a signature τ that has at least one constant symbol and a τ -formula F (that does not mention equality) such that F is satisfiable but does not have a Herbrand model.
4. A closed formula is in the class $\exists^*\forall^*$ if it has the form $\exists x_1 \dots \exists x_m \forall y_1 \dots \forall y_n F$, where F is quantifier-free and $m, n \geq 0$.
- (a) Prove that if an $\exists^*\forall^*$ -formula over a signature with no function symbols has a model then it has a finite model.
 - (b) Suggest an algorithm for deciding whether a given $\exists^*\forall^*$ -formula over a signature with no function symbols has a model.
 - (c) Argue that the satisfiability problem for the class of $\forall^*$ -formulas that may mention function symbols is undecidable.
5. Execute ground resolution to show that the following formula is unsatisfiable:

$$\forall x \forall y ((P(x) \wedge \neg Q(y, y)) \rightarrow Q(x, y)) \wedge \neg \exists x (P(x) \wedge \exists y (Q(y, y) \wedge Q(x, y))) \wedge \exists y (P(y))$$